

Anisotropic Stability for diffusions

Abstract

This note records a refinement of the one-sided Lipschitz stability estimate for generator matching. The scalar estimate propagates all drift errors with the largest eigenvalue of the symmetric Jacobian of the learnt drift. This is sharp only in an isotropic worst-case sense. In the additive-diffusion setting, the exact dual argument yields a stronger matrix-valued bound: the drift error is propagated by the derivative flow of the learnt dynamics. This matrix identifies which directions are expanded and which directions are damped. Specializing to Gaussian stochastic interpolants, and then to target distributions supported on a smooth embedded manifold, shows that the terminal score drift has a strongly contracting normal block of size comparable to $(1 - t)^{-1}$, whereas the tangential block is much less singular. Thus normal score-aligned errors are dynamically damped even when the scalar one-sided Lipschitz constant is positive because of tangential directions.

Contents

1	General idea	2
1.1	The learned flow as a terminal magnifying glass	2
1.2	The telescopic identity along characteristics	3
1.3	A coupling proof of the Wasserstein estimate	3
1.4	Recovering the scalar one-sided Lipschitz estimate	4
1.5	A two-direction picture	4
1.6	A flat gaussian example	5
1.7	A curved example: a fat torus	6
2	General anisotropic stability for velocity–diffusion generators	8
2.1	Setup	8
2.2	Derivative flow and pullback metric	8
2.3	The anisotropic stability theorem	9
2.4	A tangent–normal corollary	10
3	Specialization to Gaussian stochastic interpolants	12
3.1	The Gaussian interpolant and its exact drift	12
3.2	Posterior covariance formula for the Jacobian	12
3.3	Linear Gaussian interpolants	13
4	Manifold terminal laws	14
4.1	Geometric assumptions and notation	14
4.2	Small-noise expansion of the score	15
4.3	The moving ridge manifold	16
4.4	Eigenstructure of the score drift near the manifold	17
4.5	Anisotropic manifold stability	18
4.6	Consequences for score-aligned errors	19

5	Intrinsic anisotropic regularity, not a warped loss	20
5.1	Hölder geometry and density assumptions	20
5.2	A regularized intrinsic tangential score	21
5.3	Anisotropic Hölder expansion of the score	21
5.4	The stability certificate induced by the expansion	22
5.5	Density regularity, curvature, and expected difficulty	23

1 General idea

Before treating velocity–diffusion dynamics, it is useful to isolate the core deterministic mechanism. In this section there is no diffusion and no infinitesimal-generator notation. We compare two time-dependent velocity fields

$$v_t, \widehat{v}_t : \mathbb{R}^D \rightarrow \mathbb{R}^D, \quad 0 \leq t \leq T,$$

and the associated characteristic flows

$$\frac{d}{ds} \Phi_{t,s}(x) = v_s(\Phi_{t,s}(x)), \quad \frac{d}{ds} \widehat{\Phi}_{t,s}(x) = \widehat{v}_s(\widehat{\Phi}_{t,s}(x)), \quad \Phi_{t,t} = \widehat{\Phi}_{t,t} = \text{Id}. \quad (1)$$

We write

$$\delta v_t := v_t - \widehat{v}_t.$$

The velocity fields are assumed smooth enough in space and integrable in time so that these flows are globally defined and differentiable with respect to their initial condition. The notation in this preliminary section is deliberately different from the later drift notation: in the pure-velocity case one may read v_t as the drift a_t with diffusion coefficient equal to zero.

1.1 The learned flow as a terminal magnifying glass

The derivative of the learned flow from time t to time T is denoted by

$$\widehat{J}_{t,T}(x) := D_x \widehat{\Phi}_{t,T}(x). \quad (2)$$

It solves the linearized characteristic equation

$$\frac{d}{ds} \widehat{J}_{t,s}(x) = \nabla \widehat{v}_s(\widehat{\Phi}_{t,s}(x)) \widehat{J}_{t,s}(x), \quad \widehat{J}_{t,t}(x) = \text{Id}_D. \quad (3)$$

Thus $\widehat{J}_{t,T}(x)h$ is the terminal displacement produced by an infinitesimal displacement h inserted at time t and position x , then transported by the learned dynamics until time T .

The associated deterministic amplification matrix is

$$G_{t,T}^{\widehat{v}}(x) := \widehat{J}_{t,T}(x)^\top \widehat{J}_{t,T}(x). \quad (4)$$

It is simply the pullback of the Euclidean metric at terminal time by the learned flow. For every vector $h \in \mathbb{R}^D$,

$$\|\widehat{J}_{t,T}(x)h\|^2 = \langle h, G_{t,T}^{\widehat{v}}(x)h \rangle. \quad (5)$$

The matrix $G_{t,T}^{\widehat{v}}(x)$ is best understood as the terminal amplification quadratic form. A perturbation $h \in T_x \mathbb{R}^D \simeq \mathbb{R}^D$ made at time t is transported by the linearized learnt flow into $J_{t,T}^x h$ at time T . Therefore

$$\langle h, G_{t,T}^{\widehat{v}}(x)h \rangle = \|J_{t,T}^x h\|^2.$$

Equivalently, $G_{t,T}^{\widehat{v}}(x)$ is the Euclidean metric at time T pulled back to the time- t tangent space by the learnt flow. Thus G lives at time t , but it measures lengths after transport to time T . The

image ellipsoid itself lives at time T and is described, in terminal coordinates, by $J_{t,T}^x (J_{t,T}^x)^\top$; this is a different object from the quadratic form needed in the stability estimate.

In the deterministic pure-velocity case there is therefore no Cauchy–Schwarz loss in passing from the derivative flow to the matrix $G_{t,T}^{\hat{v}}$: the matrix records the exact squared terminal amplification of each direction.

1.2 The telescopic identity along characteristics

The key identity is a one-line chain-rule computation. Fix an initial point x_0 , and let

$$X_t := \Phi_{0,t}(x_0)$$

be the true characteristic. Consider the point obtained by following the true dynamics up to time t , and then switching to the learned dynamics from t to T :

$$F(t) := \hat{\Phi}_{t,T}(X_t).$$

At $t = 0$, this gives the learned terminal point $F(0) = \hat{\Phi}_{0,T}(x_0)$. At $t = T$, it gives the true terminal point $F(T) = \Phi_{0,T}(x_0)$. Differentiating $F(t)$ in t yields

$$\frac{d}{dt}F(t) = \partial_t \hat{\Phi}_{t,T}(X_t) + D\hat{\Phi}_{t,T}(X_t) \frac{d}{dt}X_t.$$

The first term is the derivative of the learned flow with respect to its starting time. Since starting the learned flow a little later removes the short learned displacement $\hat{v}_t$, one has

$$\partial_t \hat{\Phi}_{t,T}(x) = -D\hat{\Phi}_{t,T}(x) \hat{v}_t(x). \quad (6)$$

Together with $\dot{X}_t = v_t(X_t)$, this gives

$$\frac{d}{dt}F(t) = \hat{J}_{t,T}(X_t)(v_t(X_t) - \hat{v}_t(X_t)) = \hat{J}_{t,T}(X_t)\delta v_t(X_t).$$

Integrating from 0 to T gives the exact identity

$$\Phi_{0,T}(x_0) - \hat{\Phi}_{0,T}(x_0) = \int_0^T \hat{J}_{t,T}(\Phi_{0,t}(x_0)) \delta v_t(\Phi_{0,t}(x_0)) dt. \quad (7)$$

This is the elementary heart of the anisotropic stability estimate. The velocity error made at time t is not propagated by a scalar worst-case constant. It is propagated by the derivative of the learned flow in the actual direction of the error.

1.3 A coupling proof of the Wasserstein estimate

Let $X_0 \sim \rho_0$, and define

$$X_t := \Phi_{0,t}(X_0), \quad \hat{X}_t := \hat{\Phi}_{0,t}(X_0).$$

The terminal laws are

$$\rho_T := \mathcal{L}(X_T), \quad \hat{\rho}_T := \mathcal{L}(\hat{X}_T).$$

Using the same initial random point X_0 couples ρ_T and $\hat{\rho}_T$. By the definition of W_1 through couplings and by (7),

$$\begin{aligned} W_1(\rho_T, \hat{\rho}_T) &\leq \mathbb{E} \|X_T - \hat{X}_T\| \\ &\leq \int_0^T \mathbb{E} \left[\left\| \hat{J}_{t,T}(X_t) \delta v_t(X_t) \right\| \right] dt. \end{aligned} \quad (8)$$

Since $X_t \sim \rho_t$, this can be written as

$$W_1(\rho_T, \hat{\rho}_T) \leq \int_0^T \int_{\mathbb{R}^D} \left\| \hat{J}_{t,T}(x) \delta v_t(x) \right\| \rho_t(dx) dt. \quad (9)$$

Equivalently, by (5),

$$W_1(\rho_T, \hat{\rho}_T) \leq \int_0^T \int_{\mathbb{R}^D} \langle \delta v_t(x), G_{t,T}^{\hat{v}}(x) \delta v_t(x) \rangle^{1/2} \rho_t(dx) dt. \quad (10)$$

This is the deterministic version of the matrix-valued stability estimate. The proof has used only characteristics, the chain rule, and one explicit coupling.

Remark 1.1 (What the coupling does and does not prove). The coupling above is not claimed to be optimal. It is only a convenient coupling, obtained by running the true and learned flows from the same initial point. Its cost is already sharp enough to reveal the anisotropic propagation mechanism. In the later velocity–diffusion setting, the same idea survives, but the stochastic flow forces one to average the pullback metric over the noise.

1.4 Recovering the scalar one-sided Lipschitz estimate

The usual scalar estimate is obtained by discarding the direction of δv_t . Let

$$\hat{S}_t(x) := \frac{\nabla \hat{v}_t(x) + \nabla \hat{v}_t(x)^\top}{2},$$

and assume that

$$\lambda_{\max}(\hat{S}_t(x)) \leq \ell_t \quad \text{for all } x. \quad (11)$$

For $z_s = \hat{J}_{t,s}(x)h$,

$$\frac{d}{ds} \|z_s\|^2 = 2 \langle z_s, \hat{S}_s(\hat{\Phi}_{t,s}(x)) z_s \rangle \leq 2\ell_s \|z_s\|^2.$$

Hence

$$\|\hat{J}_{t,T}(x)h\| \leq \exp\left(\int_t^T \ell_s ds\right) \|h\|. \quad (12)$$

Inserted into (9), this gives

$$W_1(\rho_T, \hat{\rho}_T) \leq \int_0^T \exp\left(\int_t^T \ell_s ds\right) \|v_t - \hat{v}_t\|_{L^1(\rho_t)} dt. \quad (13)$$

Thus the scalar one-sided Lipschitz argument is just the isotropic domination

$$G_{t,T}^{\hat{v}}(x) \preceq \exp\left(2 \int_t^T \ell_s ds\right) \text{Id}_D.$$

It is correct, but it forgets which directions are expanded and which directions are contracted.

1.5 A two-direction picture

The simplest mental model is a fixed splitting

$$\mathbb{R}^D = E_{\parallel} \oplus E_{\perp},$$

where $E_{\parallel}$ is tangent-like and $E_{\perp}$ is normal-like. Suppose, for illustration, that the learned velocity is linear and block diagonal:

$$\hat{v}_t(x_{\parallel}, x_{\perp}) = (\ell_t x_{\parallel}, -\gamma_t x_{\perp}), \quad \gamma_t \geq 0. \quad (14)$$

Then

$$\widehat{J}_{t,T} = \begin{pmatrix} \exp\left(\int_t^T \ell_s \, ds\right) I_{\parallel} & 0 \\ 0 & \exp\left(-\int_t^T \gamma_s \, ds\right) I_{\perp} \end{pmatrix},$$

and therefore

$$G_{t,T}^{\widehat{v}} = \begin{pmatrix} \exp\left(2 \int_t^T \ell_s \, ds\right) I_{\parallel} & 0 \\ 0 & \exp\left(-2 \int_t^T \gamma_s \, ds\right) I_{\perp} \end{pmatrix}. \quad (15)$$

A tangent error is multiplied by the tangent expansion factor. A normal error is multiplied by the normal contraction factor. The scalar one-sided Lipschitz constant only sees $\max\{\ell_t, -\gamma_t\} = \ell_t$ when $\ell_t \geq 0$, and would therefore propagate both components as if they were tangent.

1.6 A flat gaussian example

We now give a flat model of the terminal-manifold situation in which the tangent and normal components can be read off exactly. The example is non-compact only to keep the formulas closed form; it should be viewed as the local model of a smooth embedded manifold in normal coordinates.

Let $D = 2$, let

$$M = \mathbb{R} \times \{0\}, \quad Z = (Y, 0), \quad Y \sim \rho(y) \, dy, \quad \rho(y) = (2\pi)^{-1/2} e^{-y^2/2},$$

and consider the pure flow-matching interpolant

$$X_t^{\circ} = tZ + \sigma_t \xi, \quad \sigma_t = 1 - t, \quad \xi \sim \mathcal{N}(0, I_2). \quad (16)$$

Writing $x = (x_{\parallel}, x_{\perp})$, the marginal density is a product Gaussian,

$$p_t(x_{\parallel}, x_{\perp}) = \frac{1}{2\pi\sigma_t\sqrt{t^2 + \sigma_t^2}} \exp\left(-\frac{x_{\parallel}^2}{2(t^2 + \sigma_t^2)} - \frac{x_{\perp}^2}{2\sigma_t^2}\right),$$

so its score is

$$s_t(x_{\parallel}, x_{\perp}) = \left(-\frac{x_{\parallel}}{t^2 + \sigma_t^2}, -\frac{x_{\perp}}{\sigma_t^2}\right). \quad (17)$$

The exact pure flow-matching velocity for this linear Gaussian interpolant is

$$v_t(x) = \frac{x}{t} + c_t s_t(x), \quad c_t = \frac{\sigma_t}{t}. \quad (18)$$

Consequently

$$v_t(x_{\parallel}, x_{\perp}) = (\lambda_{\parallel}(t)x_{\parallel}, \lambda_{\perp}(t)x_{\perp}), \quad (19)$$

with

$$\lambda_{\parallel}(t) = \frac{1}{t} - \frac{1-t}{t(t^2 + (1-t)^2)}, \quad \lambda_{\perp}(t) = \frac{1}{t} - \frac{1}{t(1-t)} = -\frac{1}{1-t}. \quad (20)$$

The normal eigenvalue is singular and negative. The tangent eigenvalue remains bounded near terminal time; in fact $\lambda_{\parallel}(t) \rightarrow 1$ as $t \uparrow 1$. If the learned velocity has the same block structure, the normal derivative-flow factor is

$$\exp\left(\int_t^T \lambda_{\perp}(s) \, ds\right) = \frac{1-T}{1-t}. \quad (21)$$

Thus a normal velocity error made at time t is damped by the factor $(1-T)/(1-t)$ before terminal time T . The scalar one-sided Lipschitz estimate sees only the largest eigenvalue, which is tangent, and misses this damping.

The same example also explains the phrase “score errors are mostly normal” near a terminal manifold. On a typical sample from p_t , one has $|x_\perp| \asymp \sigma_t$ and $|x_\parallel| = O(1)$. Therefore (17) gives

$$|s_t^\perp(x)| \asymp \sigma_t^{-1}, \quad |s_t^\parallel(x)| = O(1) \quad (t \uparrow 1).$$

The density $\rho(y) = (2\pi)^{-1/2} e^{-y^2/2}$ was chosen so that a genuine tangential score is present. It remains much smaller than the normal restoring score. If a score error is mainly a relative error in the normal restoring force, then the matrix stability bound propagates it with the damping factor (21). If instead the error comes from estimating the tangential density ρ , then the error is tangent and the bound correctly treats it as persistent.

This flat calculation is the local template for the manifold case. Curvature does not change the leading normal mechanism; it shifts the high-density ridge by a lower-order amount and rotates the tangent–normal frame. The full anisotropic theory below is essentially this elementary picture with moving projectors, curvature remainders, and, later, additive diffusion.

1.7 A curved example: a fat torus

The flat model shows the leading normal contraction, but it hides the lower-order geometric terms. A simple curved model where these terms can be seen explicitly is a torus of revolution in $\mathbb{R}^3$. We choose a *fat* embedded torus so that both the Gaussian curvature and the mean-curvature scalar change sign.

Let $0 < r < R < 2r$, and write

$$\Theta(\varphi, \theta) = ((R + r \cos \theta) \cos \varphi, (R + r \cos \theta) \sin \varphi, r \sin \theta), \quad (\varphi, \theta) \in \mathbb{T}^2. \quad (22)$$

The outward unit normal is

$$N(\varphi, \theta) = (\cos \theta \cos \varphi, \cos \theta \sin \varphi, \sin \theta). \quad (23)$$

With the convention used later in the manuscript, namely that the round sphere with outward normal has inward mean-curvature vector, the two principal curvatures of the torus are

$$\kappa_\theta = -\frac{1}{r}, \quad \kappa_\varphi = -\frac{\cos \theta}{R + r \cos \theta}. \quad (24)$$

Thus the Gaussian curvature is

$$K(\theta) = \kappa_\theta \kappa_\varphi = \frac{\cos \theta}{r(R + r \cos \theta)}. \quad (25)$$

It is positive on the outside of the torus, where $\cos \theta > 0$, and negative on the inside, where $\cos \theta < 0$. The scalar trace of the second fundamental form is

$$h(\theta) := \kappa_\theta + \kappa_\varphi = -\frac{R + 2r \cos \theta}{r(R + r \cos \theta)}, \quad H(\varphi, \theta) = h(\theta)N(\varphi, \theta). \quad (26)$$

Since we chose $R < 2r$, this mean-curvature scalar also changes sign. For example, with $r = 1$ and $R = 3/2$,

$$h(0) < 0, \quad h(\pi) > 0, \quad h(\theta) = 0 \iff \cos \theta = -\frac{3}{4}.$$

This is a useful warning: the sign-changing curvature visible in the torus is not only an aesthetic feature. The Gaussian curvature changes sign at $\cos \theta = 0$, while the ridge displacement below is driven by the mean curvature h , and therefore changes sign at a different latitude.

Let the terminal law be the normalized area measure on this torus, and consider again the pure flow-matching interpolant

$$X_t^\circ = tZ + \sigma_t \xi, \quad \sigma_t = 1 - t, \quad Z \sim \frac{\text{vol}_M}{\text{vol}_M(M)}, \quad \xi \sim \mathcal{N}(0, I_3). \quad (27)$$

The scaled torus is $M_t = tM$. In the tube of M_t , write a point as

$$x = t\Theta(\varphi, \theta) + \zeta N(\varphi, \theta), \quad |\zeta| \leq L\sigma_t. \quad (28)$$

The principal curvatures and mean curvature of M_t are scaled by t^{-1} :

$$\kappa_i^{(t)} = t^{-1}\kappa_i, \quad h_t = t^{-1}h. \quad (29)$$

A direct Laplace expansion of the Gaussian convolution in the normal coordinate gives

$$s_t(x) = \left(-\frac{\zeta}{\sigma_t^2} + \frac{h(\theta)}{2t} \right) N(\varphi, \theta) + \frac{\zeta}{2} \nabla_{M_t} h_t(\varphi, \theta) + \text{Rem}_t(\varphi, \theta, \zeta), \quad (30)$$

where, for fixed L , the remainder is lower order uniformly in the tube. The first term is the universal normal Gaussian restoring score. The second normal term is the mean-curvature correction. The tangential term is caused by the variation of mean curvature along the torus; it vanishes on the symmetry circles $\theta = 0$ and $\theta = \pi$, and is of size $O(|\zeta|)$ in the typical tube.

The normal ridge is obtained by setting the leading normal score to zero. Hence

$$\mathcal{R}_t = \left\{ t\Theta(\varphi, \theta) + \frac{\sigma_t^2}{2t} h(\theta) N(\varphi, \theta) + O(\sigma_t^3) : (\varphi, \theta) \in \mathbb{T}^2 \right\}. \quad (31)$$

Thus curvature does not merely create an abstract correction term: it visibly moves the high-density ridge. On the outer equator $\theta = 0$, one has $h(0) < 0$, so the ridge is shifted opposite to the outward normal. On the inner equator $\theta = \pi$, for the fat torus $R < 2r$, one has $h(\pi) > 0$, so the ridge is shifted in the outward-normal direction, namely toward the hole of the torus. At the latitude where $h = 0$, the first curvature displacement of the ridge vanishes even though the Gaussian curvature is still negative.

The exact pure flow-matching velocity is again

$$v_t(x) = \frac{x}{t} + \frac{\sigma_t}{t} s_t(x). \quad (32)$$

Taking the normal component in the coordinates (28) gives, to leading order,

$$\langle v_t(x), N(\varphi, \theta) \rangle = -\frac{\zeta}{1-t} + \frac{\sigma_t}{2t^2} h(\theta) + O_L(\sigma_t). \quad (33)$$

Consequently the normal linearized eigenvalue is still

$$\partial_\zeta \langle v_t, N \rangle = -\frac{1}{1-t} + O(1). \quad (34)$$

This is exactly the same terminal contraction as in the flat model. Curvature shifts the center of the restoring field, but it does not change its leading slope.

The tangential directions remember the curvature signs. If e_i is a principal direction, then the tangential block of the score Hessian contains the standard parallel-surface term

$$\langle e_i, \nabla s_t(x) e_i \rangle = \frac{\zeta \kappa_i^{(t)}}{\sigma_t^2 (1 - \zeta \kappa_i^{(t)})} + O_L(1). \quad (35)$$

Therefore the corresponding tangential eigenvalue of ∇v_t is

$$\lambda_i^\parallel(t, x) = \frac{1}{t} + \frac{\sigma_t}{t} \frac{\zeta \kappa_i^{(t)}}{\sigma_t^2 (1 - \zeta \kappa_i^{(t)})} + O_L(\sigma_t). \quad (36)$$

On the typical shell $\zeta = \sigma_t \nu$, this remains of order one and its sign depends on the product $\nu \kappa_i$. In particular, the azimuthal curvature κ_φ changes sign between the outside and the inside of the

torus, so this tangential curvature correction may be expansive in one region and contractive in another. On the ridge itself, where $\zeta = O(\sigma_t^2)$, the curvature contribution in (36) is even smaller, of order σ_t .

This example isolates the role of curvature in the anisotropic estimate. The normal block is universally dominated by the Gaussian restoring slope $-(1-t)^{-1}$. The mean curvature moves the ridge by $\sigma_t^2 H_t/2$. The signed principal curvatures enter the tangential block through bounded parallel-surface corrections. Thus the scalar one-sided Lipschitz constant may be affected by the largest tangential curvature contribution, while a normal velocity error is still propagated by the strongly contracting normal derivative flow.

2 General anisotropic stability for velocity–diffusion generators

Throughout this section the state space is $\mathbb{R}^D$, the final time is denoted by $T \leq 1$, and the two dynamics have the same additive diffusion coefficient. This is the setting most directly used by flow matching and score-based generator matching when the diffusion schedule is chosen by the modeller and only the drift is learnt.

2.1 Setup

Let

$$L_t \varphi(x) = \langle a_t(x), \nabla \varphi(x) \rangle + b_t^2 \Delta \varphi(x), \quad \widehat{L}_t \varphi(x) = \langle \widehat{a}_t(x), \nabla \varphi(x) \rangle + b_t^2 \Delta \varphi(x),$$

where $b_t \geq 0$ depends only on time. Let $(p_t)_{0 \leq t \leq T}$ and $(\widehat{p}_t)_{0 \leq t \leq T}$ solve the weak Kolmogorov forward equations associated with L_t and $\widehat{L}_t$, with common initial law

$$p_0 = \widehat{p}_0.$$

For a bounded terminal test function f , write

$$\widehat{P}_{t,T} f(x) := \mathbb{E}[f(\widehat{X}_T^{t,x})],$$

where

$$d\widehat{X}_s^{t,x} = \widehat{a}_s(\widehat{X}_s^{t,x}) ds + \sqrt{2}b_s dB_s, \quad \widehat{X}_t^{t,x} = x. \quad (37)$$

Then $\phi_t = \widehat{P}_{t,T} f$ solves the backward equation

$$\partial_t \phi_t + \widehat{L}_t \phi_t = 0, \quad \phi_T = f.$$

Assumption 2.1 (Regularity for the stability argument). *We assume enough regularity to justify the computations below. For instance, it is sufficient that a_t and $\widehat{a}_t$ are locally Lipschitz in space, have at most linear growth uniformly on compact time intervals, $\widehat{a}_t$ is C^1 in space, $b \in L^2(0, T)$, and the learnt SDE admits a non-explosive stochastic flow differentiable with respect to its initial condition. The estimates then extend from smooth compactly supported terminal functions to bounded Lipschitz functions by standard smoothing and truncation.*

2.2 Derivative flow and pullback metric

Let $J_{t,s}^x$ denote the derivative of the learnt stochastic flow with respect to its initial condition:

$$J_{t,s}^x := D_x \widehat{X}_s^{t,x}. \quad (38)$$

Since the diffusion is additive, $J_{t,s}^x$ solves the random linear equation

$$\partial_s J_{t,s}^x = \nabla \widehat{a}_s(\widehat{X}_s^{t,x}) J_{t,s}^x, \quad J_{t,t}^x = \text{Id}_D. \quad (39)$$

The additive-noise assumption is used exactly here: the Brownian term has no derivative with respect to the initial condition. If the diffusion matrix were state-dependent, additional derivative terms would appear.

For each (t, x) , define the symmetric positive semidefinite matrix

$$G_{t,T}^{\hat{a}}(x) := \mathbb{E} \left[(J_{t,T}^x)^\top J_{t,T}^x \right]. \quad (40)$$

This is the *local amplification metric* of the learnt dynamics between times t and T . If $b \equiv 0$, then the flow is deterministic and

$$G_{t,T}^{\hat{a}}(x) = (D\hat{\Phi}_{t,T}(x))^\top D\hat{\Phi}_{t,T}(x),$$

which is precisely the pullback of the Euclidean metric at final time by the learnt flow. Its eigenvectors are the input directions at (t, x) , and its eigenvalues are the squared stretching factors of those directions. With additive diffusion, the same pullback metric is averaged over the Brownian trajectories.

The matrix $G_{t,T}^{\hat{a}}$ is therefore the anisotropic object hidden behind the usual scalar one-sided Lipschitz estimate. Indeed, if

$$S_t(x) := \frac{\nabla \hat{a}_t(x) + \nabla \hat{a}_t(x)^\top}{2}$$

satisfies

$$\lambda_{\max}(S_t(x)) \leq \ell_t \quad \text{for all } x,$$

then, for every vector v ,

$$\|J_{t,T}^x v\|^2 \leq \exp \left(2 \int_t^T \ell_s \, ds \right) \|v\|^2,$$

and hence

$$G_{t,T}^{\hat{a}}(x) \preceq \exp \left(2 \int_t^T \ell_s \, ds \right) \text{Id}_D. \quad (41)$$

The scalar bound is just the Loewner domination (41); it forgets the eigenstructure of $G_{t,T}^{\hat{a}}$.

2.3 The anisotropic stability theorem

Theorem 2.2 (Anisotropic W_1 stability for additive diffusions). *Assume Assumption 2.1. Let*

$$\delta a_t(x) := a_t(x) - \hat{a}_t(x).$$

Then

$$W_1(p_T, \hat{p}_T) \leq \int_0^T \int_{\mathbb{R}^D} \mathbb{E} [\|J_{t,T}^x \delta a_t(x)\|] p_t(dx) \, dt. \quad (42)$$

Consequently,

$$W_1(p_T, \hat{p}_T) \leq \int_0^T \int_{\mathbb{R}^D} \left\langle \delta a_t(x), G_{t,T}^{\hat{a}}(x) \delta a_t(x) \right\rangle^{1/2} p_t(dx) \, dt. \quad (43)$$

In particular, if $\lambda_{\max}(S_t(x)) \leq \ell_t$, then (43) implies the usual scalar one-sided Lipschitz bound

$$W_1(p_T, \hat{p}_T) \leq \int_0^T \exp \left(\int_t^T \ell_s \, ds \right) \|a_t - \hat{a}_t\|_{L^1(p_t)} \, dt. \quad (44)$$

Proof. Let $f \in C_c^\infty(\mathbb{R}^D)$ satisfy $\|\nabla f\|_\infty \leq 1$, and set

$$\phi_t(x) := \widehat{P}_{t,T} f(x).$$

The duality identity for generator errors gives

$$\langle p_T - \widehat{p}_T, f \rangle = \int_0^T \int_{\mathbb{R}^D} \langle \delta a_t(x), \nabla \phi_t(x) \rangle p_t(dx) dt. \quad (45)$$

The gradient of ϕ_t is represented through the derivative flow. Indeed, differentiating $\phi_t(x) = \mathbb{E}[f(\widehat{X}_T^{t,x})]$ with respect to x gives

$$\nabla \phi_t(x) = \mathbb{E} \left[(J_{t,T}^x)^\top \nabla f(\widehat{X}_T^{t,x}) \right]. \quad (46)$$

Therefore, pointwise in (t, x) ,

$$\begin{aligned} |\langle \delta a_t(x), \nabla \phi_t(x) \rangle| &= \left| \mathbb{E} \left[\left\langle J_{t,T}^x \delta a_t(x), \nabla f(\widehat{X}_T^{t,x}) \right\rangle \right] \right| \\ &\leq \mathbb{E} \left[\|J_{t,T}^x \delta a_t(x)\| \left\| \nabla f(\widehat{X}_T^{t,x}) \right\| \right] \\ &\leq \mathbb{E} \left[\|J_{t,T}^x \delta a_t(x)\| \right]. \end{aligned}$$

Substitution in (45) yields (42). Then Jensen or Cauchy–Schwarz gives

$$\mathbb{E} \left[\|J_{t,T}^x \delta a_t(x)\| \right] \leq \left(\mathbb{E} \left[\|J_{t,T}^x \delta a_t(x)\|^2 \right] \right)^{1/2} = \left\langle \delta a_t(x), G_{t,T}^{\widehat{a}}(x) \delta a_t(x) \right\rangle^{1/2},$$

which proves (43).

Finally, if $S_t \preceq \ell_t \text{Id}$, then, for every deterministic vector v , the process $Z_s = J_{t,s}^x v$ satisfies

$$\frac{d}{ds} \|Z_s\|^2 = 2 \langle Z_s, S_s(\widehat{X}_s^{t,x}) Z_s \rangle \leq 2\ell_s \|Z_s\|^2.$$

Gronwall's lemma gives

$$\|J_{t,T}^x v\| \leq \exp \left(\int_t^T \ell_s ds \right) \|v\|,$$

which inserted into (42) gives (44). Taking the supremum over 1-Lipschitz terminal functions and using Kantorovich–Rubinstein duality concludes the proof. $\square$

Remark 2.3 (Why this is sharper). The scalar one-sided Lipschitz constant is

$$\ell_t = \text{ess sup}_x \lambda_{\max} \left(\frac{\nabla \widehat{a}_t(x) + \nabla \widehat{a}_t(x)^\top}{2} \right).$$

It only sees the most expansive direction. The matrix $G_{t,T}^{\widehat{a}}(x)$, in contrast, sees the direction of the actual error $\delta a_t(x)$. If $\delta a_t(x)$ lies mostly in directions contracted by the learnt flow, then $\langle \delta a, G \delta a \rangle^{1/2}$ can be much smaller than $\exp(\int \ell) \|\delta a\|$.

2.4 A tangent–normal corollary

The previous theorem becomes useful when the symmetric Jacobian has a distinguished splitting. The following statement is deliberately formulated at the level of a moving splitting along the learnt characteristics; it is the right form for manifold terminal laws, where tangent and normal spaces move in space and time.

Let $P_s = P_s(\widehat{X}_s^{t,x})$ be an orthogonal projector and $Q_s = \text{Id} - P_s$. Think of P_s as a tangential projector and Q_s as a normal projector. Define

$$S_s = \frac{\nabla \widehat{a}_s(\widehat{X}_s^{t,x}) + \nabla \widehat{a}_s(\widehat{X}_s^{t,x})^\top}{2}.$$

Proposition 2.4 (Two-block anisotropic propagation). *Assume that along the learnt trajectories one has, for deterministic integrable functions $\ell_s^\parallel, \gamma_s^\perp \geq 0$, and $\kappa_s \geq 0$,*

$$\langle u, S_s u \rangle \leq \ell_s^\parallel \|u\|^2 \quad \text{for all } u \in \text{Im}(P_s), \quad (47)$$

$$\langle w, S_s w \rangle \leq -\gamma_s^\perp \|w\|^2 \quad \text{for all } w \in \text{Im}(Q_s), \quad (48)$$

and that the off-diagonal and moving-frame terms are bounded by κ_s in the sense that, for $z_s = J_{t,s}^x v$,

$$\left| \frac{d}{ds} \|P_s z_s\| \right| \leq \ell_s^\parallel \|P_s z_s\| + \kappa_s \|Q_s z_s\| \quad \text{whenever } \|P_s z_s\| > 0, \quad (49)$$

$$\left| \frac{d}{ds} \|Q_s z_s\| \right| \leq \kappa_s \|P_s z_s\| - \gamma_s^\perp \|Q_s z_s\| \quad \text{whenever } \|Q_s z_s\| > 0, \quad (50)$$

with the inequalities understood in the upper Dini sense at zeros. Let $R_{t,s} \in \mathbb{R}^{2 \times 2}$ solve

$$\partial_s R_{t,s} = \begin{pmatrix} \ell_s^\parallel & \kappa_s \\ \kappa_s & -\gamma_s^\perp \end{pmatrix} R_{t,s}, \quad R_{t,t} = I_2. \quad (51)$$

Then

$$\|J_{t,T}^x v\| \leq (1, 1) R_{t,T} \begin{pmatrix} \|P_t v\| \\ \|Q_t v\| \end{pmatrix}. \quad (52)$$

Consequently,

$$W_1(p_T, \widehat{p}_T) \leq \int_0^T \int_{\mathbb{R}^D} (1, 1) R_{t,T} \begin{pmatrix} \|P_t \delta a_t(x)\| \\ \|Q_t \delta a_t(x)\| \end{pmatrix} p_t(dx) dt. \quad (53)$$

If $\kappa \equiv 0$, this reduces to

$$W_1(p_T, \widehat{p}_T) \leq \int_0^T \int_{\mathbb{R}^D} \left[\exp \left(\int_t^T \ell_s^\parallel ds \right) \|P_t \delta a_t(x)\| + \exp \left(- \int_t^T \gamma_s^\perp ds \right) \|Q_t \delta a_t(x)\| \right] p_t(dx) dt. \quad (54)$$

Proof. Let

$$u_s^\parallel := \|P_s z_s\|, \quad \nu_s^\perp := \|Q_s z_s\|.$$

Assumption (49) says exactly that the vector $\nu_s = (\nu_s^\parallel, \nu_s^\perp)^\top$ is a subsolution, in the comparison sense, of the linear cooperative system

$$\partial_s r_s = \begin{pmatrix} \ell_s^\parallel & \kappa_s \\ \kappa_s & -\gamma_s^\perp \end{pmatrix} r_s, \quad r_t = \nu_t.$$

Since the off-diagonal entries are nonnegative, the system is order-preserving. Hence

$$\nu_s \leq R_{t,s} \nu_t$$

componentwise. Since $\|z_T\| \leq \nu_T^\parallel + \nu_T^\perp$, (52) follows. Inserting (52) into Theorem 2.2 yields (53). When $\kappa = 0$, the two-dimensional propagator is diagonal and gives (54). $\square$

Remark 2.5 (What is hidden in κ_s). If P_s depends on space and time, then κ_s contains two effects: the genuine off-diagonal blocks $P_s(\nabla \widehat{a}_s)Q_s$ and $Q_s(\nabla \widehat{a}_s)P_s$, and the rotation of the frame, through terms such as $\dot{P}_s Q_s$ along the trajectory. In the manifold case, these terms are controlled by curvature and by the speed at which the relevant ridge moves.

3 Specialization to Gaussian stochastic interpolants

We now specialize the previous estimate to the Euclidean stochastic interpolants used in score-based generator matching and flow matching.

3.1 The Gaussian interpolant and its exact drift

Let $Z \sim p^\star$ be the target random variable in $\mathbb{R}^D$, let $\xi \sim \mathcal{N}(0, I_D)$ be independent, and set

$$X_t^\circ = m_t(Z) + \sigma_t \xi, \quad 0 < t < T < 1, \quad (55)$$

where $m_t : \mathbb{R}^D \rightarrow \mathbb{R}^D$ is a smooth deterministic map and $\sigma_t > 0$. The conditional law of X_t° given $Z = z$ is

$$p_t(\mathrm{d}x \mid z) = \mathcal{N}(m_t(z), \sigma_t^2 I_D)(\mathrm{d}x).$$

Let p_t be the marginal law of X_t° , and let

$$s_t(x) := \nabla \log p_t(x)$$

be its score, whenever the density is positive and smooth.

The conditional deterministic velocity transporting the conditional Gaussian path is

$$u_t(x \mid z) = \dot{m}_t(z) + \frac{\dot{\sigma}_t}{\sigma_t}(x - m_t(z)).$$

If we choose an additive generative diffusion $b_t \geq 0$, then the corresponding conditional SDE drift preserving the same conditional marginals is

$$a_t(x \mid z) = u_t(x \mid z) + b_t^2 \nabla_x \log p_t(x \mid z) = \dot{m}_t(z) + \eta_t(x - m_t(z)),$$

where

$$\eta_t := \frac{\dot{\sigma}_t}{\sigma_t} - \frac{b_t^2}{\sigma_t^2}. \quad (56)$$

The exact marginal drift is the posterior average

$$a_t(x) = \mathbb{E}[\dot{m}_t(Z) + \eta_t(x - m_t(Z)) \mid X_t^\circ = x]. \quad (57)$$

Equivalently,

$$a_t(x) = v_t(x) + b_t^2 s_t(x), \quad (58)$$

where

$$v_t(x) := \mathbb{E}[\nu_t(X_t^\circ \mid Z) \mid X_t^\circ = x]$$

is the probability-flow velocity.

3.2 Posterior covariance formula for the Jacobian

Let $\pi_{t,x}$ denote the posterior law of Z conditionally on $X_t^\circ = x$. Thus, formally,

$$\pi_{t,x}(\mathrm{d}z) \propto \exp\left(-\frac{\|x - m_t(z)\|^2}{2\sigma_t^2}\right) p^\star(\mathrm{d}z).$$

Define

$$h_t(z) := \dot{m}_t(z) - \eta_t m_t(z).$$

Then (57) becomes

$$a_t(x) = \eta_t x + \mathbb{E}_{\pi_{t,x}}[h_t(Z)]. \quad (59)$$

Differentiating the posterior expectation gives the covariance identity

$$\nabla a_t(x) = \eta_t I_D + \frac{1}{\sigma_t^2} \text{Cov}_{\pi_{t,x}}(h_t(Z), m_t(Z)), \quad (60)$$

where

$$\text{Cov}_{\pi_{t,x}}(U, V) := \mathbb{E}_{\pi_{t,x}} \left[(U - \mathbb{E}U)(V - \mathbb{E}V)^\top \right].$$

For a general nonlinear map m_t , this covariance matrix need not be symmetric. Its symmetric part is the object entering one-sided stability:

$$S_t(x) = \text{sym} \nabla a_t(x) = \eta_t I_D + \frac{1}{2\sigma_t^2} [\text{Cov}_{\pi_{t,x}}(h_t, m_t) + \text{Cov}_{\pi_{t,x}}(m_t, h_t)].$$

Proof of (60). For any posterior-integrable vector-valued function g , differentiating the tilted expectation gives

$$\nabla_x \mathbb{E}_{\pi_{t,x}}[g(Z)] = \frac{1}{\sigma_t^2} \text{Cov}_{\pi_{t,x}}(g(Z), m_t(Z)).$$

Indeed, the derivative of the log-likelihood in the direction $u \in \mathbb{R}^D$ is

$$\partial_u \left[-\frac{\|x - m_t(z)\|^2}{2\sigma_t^2} \right] = -\frac{\langle x - m_t(z), u \rangle}{\sigma_t^2},$$

and subtracting the derivative of the normalizing constant centers this quantity. The constant term $-x/\sigma_t^2$ cancels after centering, leaving the covariance with $m_t(Z)$. Applying this to $g = h_t$ in (59) proves the formula. $\square$

3.3 Linear Gaussian interpolants

The most important case for the discussion below is

$$m_t(z) = tz. \quad (61)$$

Then

$$X_t^\circ = tZ + \sigma_t \xi, \quad \eta_t = \frac{\dot{\sigma}_t}{\sigma_t} - \frac{b_t^2}{\sigma_t^2}.$$

The posterior mean is related to the score by Tweedie's formula

$$\mathbb{E}[tZ \mid X_t^\circ = x] = x + \sigma_t^2 s_t(x). \quad (62)$$

Using (59), one obtains the explicit score form

$$a_t(x) = \frac{x}{t} + c_t s_t(x), \quad c_t := \sigma_t^2 \left(\frac{1}{t} - \eta_t \right) = \frac{\sigma_t^2}{t} - \sigma_t \dot{\sigma}_t + b_t^2. \quad (63)$$

Therefore

$$\nabla a_t(x) = \frac{1}{t} I_D + c_t \nabla s_t(x). \quad (64)$$

Two benchmark schedules are worth keeping in mind.

(i) **Vanilla flow matching.** If $\sigma_t = 1 - t$ and $b_t = 0$, then

$$\eta_t = -\frac{1}{1-t}, \quad c_t = \frac{1-t}{t}.$$

(ii) **Rescaled diffusion.** If $\sigma_t^2 = 1 - t^2$ and $b_t^2 = t^{-1}$, then

$$\eta_t = -\frac{1+t^2}{t(1-t^2)}, \quad c_t = \frac{2}{t}.$$

For linear interpolants, the covariance formula is symmetric because $h_t(z) = (1/t - \eta_t)m_t(z)$. In fact,

$$\nabla a_t(x) = \eta_t I_D + \frac{1}{\sigma_t^2} \left(\frac{1}{t} - \eta_t \right) \text{Cov}(tZ \mid X_t^\circ = x). \quad (65)$$

Formula (64) and formula (65) are the same identity written in score form and in posterior covariance form.

4 Manifold terminal laws

We now assume that the target is supported on a smooth embedded manifold. This is the regime where the anisotropic estimate is especially informative.

4.1 Geometric assumptions and notation

Let $M \subset \mathbb{R}^D$ be a compact connected embedded C^4 submanifold of dimension d , without boundary, and with positive reach $\tau_M > 0$. Let

$$p^\star = \rho \text{vol}_M,$$

where $\rho \in C^3(M)$ is positive and bounded away from 0. Consider the linear Gaussian interpolant

$$X_t^\circ = tZ + \sigma_t \xi, \quad Z \sim \rho \text{vol}_M, \quad \xi \sim \mathcal{N}(0, I_D),$$

with $\sigma_t \downarrow 0$ as $t \uparrow 1$. Write

$$M_t := tM.$$

The reach of M_t is $t\tau_M$. The pushforward of $p^\star$ by $z \mapsto tz$ has density

$$\rho_t(ty) = t^{-d} \rho(y)$$

with respect to vol_{M_t} . Thus

$$p_t(x) = \int_{M_t} (2\pi\sigma_t^2)^{-D/2} \exp\left(-\frac{\|x - y\|^2}{2\sigma_t^2}\right) \rho_t(y) \text{vol}_{M_t}(dy). \quad (66)$$

For x in the tubular neighborhood of M_t , denote by $\pi_t(x) \in M_t$ its nearest-point projection and write

$$x = y + n, \quad y := \pi_t(x), \quad n := x - \pi_t(x) \in N_y M_t.$$

Let $P_t(x)$ and $Q_t(x)$ be the orthogonal projectors onto $T_y M_t$ and $N_y M_t$. We write II_t for the second fundamental form of M_t , with the convention that in normal coordinates around $y \in M_t$,

$$z(u) = y + u + \frac{1}{2} II_t(u, u) + O(\|u\|^3), \quad u \in T_y M_t.$$

The mean-curvature vector is

$$H_t(y) := \text{Tr}_{T_y M_t} II_t.$$

With this convention, a round sphere with outward normal has inward mean-curvature vector. Curvatures scale as

$$II_t(ty) = t^{-1} II_M(y), \quad H_t(ty) = t^{-1} H_M(y). \quad (67)$$

Finally, for a normal vector $n \in N_y M_t$, define the shape operator $A_{t,n} : T_y M_t \rightarrow T_y M_t$ by

$$\langle A_{t,n} u, v \rangle = \langle II_t(u, v), n \rangle. \quad (68)$$

4.2 Small-noise expansion of the score

The density (66) is a Gaussian convolution of a measure on M_t . The following expansion is the local Laplace calculation that explains the tangent–normal eigenstructure. It is stated in the form needed for stability; constants are uniform for t in compact subintervals of $(0, 1]$ and for $\sigma_t/(t\tau_M)$ sufficiently small.

Proposition 4.1 (Tubular expansion of the manifold score). *Fix $L > 0$. Uniformly for points $x = y + n$ with $y \in M_t$, $n \in N_y M_t$, and $\|n\| \leq L\sigma_t$, one has*

$$Q_t(x)s_t(x) = -\frac{n}{\sigma_t^2} + \frac{1}{2}H_t(y) + R_t^\perp(x), \quad (69)$$

$$P_t(x)s_t(x) = \nabla_{M_t} \log \rho_t(y) + R_t^\parallel(x), \quad (70)$$

where $\|R_t^\perp(x)\| + \|R_t^\parallel(x)\| \leq C_L(1 + \|n\|/\sigma_t)$ and, on the thinner tube $\|n\| \leq L\sigma_t^2$, the sharper bound $\|R_t^\perp(x)\| + \|R_t^\parallel(x)\| \leq C_L\sigma_t$ holds.

Moreover, the Hessian of the score has the following leading blocks. On $\|n\| \leq L\sigma_t$,

$$Q_t(x)\nabla s_t(x)Q_t(x) = -\sigma_t^{-2}Q_t(x) + O_L(1), \quad (71)$$

$$P_t(x)\nabla s_t(x)P_t(x) = \text{Hess}_{M_t} \log \rho_t(y) + \frac{1}{\sigma_t^2}A_{t,n}(I - A_{t,n})^{-1} + O_L(1 + \|n\|/\sigma_t), \quad (72)$$

$$P_t(x)\nabla s_t(x)Q_t(x) = O_L(1 + \sigma_t^{-1}\|n\|/\sigma_t), \quad Q_t(x)\nabla s_t(x)P_t(x) = O_L(1 + \sigma_t^{-1}\|n\|/\sigma_t). \quad (73)$$

On the thinner ridge tube $\|n\| \leq L\sigma_t^2$, the tangential block is only order one:

$$P_t\nabla s_t P_t = \text{Hess}_{M_t} \log \rho_t + \frac{1}{\sigma_t^2}A_{t,n} + O_L(1). \quad (74)$$

Proof sketch with the relevant computations. Work in geodesic coordinates $u \in T_y M_t$ around $y \in M_t$. Locally,

$$z(u) = y + u + \frac{1}{2}II_t(u, u) + O(\|u\|^3), \quad \text{vol}_{M_t}(dz(u)) = (1 + O(\|u\|^2)) du.$$

Since $x = y + n$ with $n \perp T_y M_t$,

$$\begin{aligned} \|x - z(u)\|^2 &= \left\| n - u - \frac{1}{2}II_t(u, u) + O(\|u\|^3) \right\|^2 \\ &= \|n\|^2 + \|u\|^2 - \langle n, II_t(u, u) \rangle + O(\|u\|^3 + \|n\|\|u\|^3 + \|u\|^4). \end{aligned}$$

After the change of variables $u = \sigma_t w$, the Gaussian factor localizes the integral to $\|w\| = O(1)$. Expanding the density $\rho_t(z(u))$, the volume element, and the exponential gives

$$\begin{aligned} p_t(y + n) &= (2\pi\sigma_t^2)^{-(D-d)/2} \exp\left(-\frac{\|n\|^2}{2\sigma_t^2}\right) \rho_t(y) \\ &\quad \times \left[1 + \frac{1}{2}\langle n, H_t(y) \rangle + O_L(\sigma_t + \|n\|^2 + \sigma_t^2) \right] \end{aligned}$$

for $\|n\| \leq L\sigma_t$, with differentiated versions obtained by differentiating under the integral. Taking $\nabla \log p_t$ gives the displayed score expansion.

The Hessian block (71) follows by differentiating the leading normal term $-n/\sigma_t^2$. For the tangential block, one uses the standard Hessian formula for the squared distance to a submanifold in a tube:

$$P_t \nabla^2 \left(\frac{1}{2} \text{dist}(\cdot, M_t)^2 \right) P_t = -A_{t,n}(I - A_{t,n})^{-1},$$

with our convention (68). Since the Gaussian contributes $-\text{dist}(\cdot, M_t)^2/(2\sigma_t^2)$ to the log-density, this gives the term $\sigma_t^{-2}A_{t,n}(I - A_{t,n})^{-1}$. The remaining terms come from differentiating the amplitude ρ_t , the curvature correction, and the projection map. The positive reach assumption keeps these derivatives uniformly controlled as long as $\|n\| < ct\tau_M$. $\square$

Remark 4.2 (Typical tube versus ridge tube). Under p_t , the normal displacement n is typically of order σ_t . Therefore the curvature contribution $\sigma_t^{-2}A_{t,n}$ in the tangential Hessian is typically of order σ_t^{-1} . On the ridge of the density, however, the normal displacement is of order σ_t^2 , and this tangential curvature contribution becomes order one. This distinction is important: the density ridge is smoother than a typical noisy sample cloud.

4.3 The moving ridge manifold

The normal ridge of p_t is the set of points in the tube where the normal component of the score vanishes:

$$\mathcal{R}_t := \{x \in \text{Tube}(M_t) : Q_t(x)s_t(x) = 0\}.$$

By (69) and the implicit function theorem, $\mathcal{R}_t$ is, for σ_t small, a normal graph over M_t . More precisely,

$$\mathcal{R}_t = \{\Gamma_t(y) := ty + r_t(y) : y \in M\}, \quad (75)$$

where $r_t(y) \in N_{ty}M_t$ satisfies

$$r_t(y) = \frac{\sigma_t^2}{2}H_t(ty) + O(\sigma_t^3) = \frac{\sigma_t^2}{2t}H_M(y) + O(\sigma_t^3). \quad (76)$$

The sign is fixed by our convention for H . For a round sphere with outward normal, H points inward, so the ridge is shifted slightly inward, as expected.

Thus the high-density manifold seen by the score is not exactly $M_t = tM$, but a nearby time-dependent ridge $\mathcal{R}_t$. It converges to M as $t \uparrow 1$, but it moves with an additional curvature-dependent velocity. Differentiating (76) gives the leading motion of the ridge labels:

$$\partial_t \Gamma_t(y) = y + \left(\frac{\sigma_t \dot{\sigma}_t}{t} - \frac{\sigma_t^2}{2t^2} \right) H_M(y) + O(\sigma_t^2 + |\dot{\sigma}_t| \sigma_t^2). \quad (77)$$

For the two benchmark schedules:

- (i) If $\sigma_t = 1 - t$, then

$$\frac{\sigma_t \dot{\sigma}_t}{t} - \frac{\sigma_t^2}{2t^2} = -\frac{1-t}{t} - \frac{(1-t)^2}{2t^2}.$$

The curvature correction to the ridge velocity vanishes as $t \uparrow 1$.

- (ii) If $\sigma_t^2 = 1 - t^2$, then

$$\frac{\sigma_t \dot{\sigma}_t}{t} - \frac{\sigma_t^2}{2t^2} = -\frac{1}{2} - \frac{1}{2t^2},$$

which converges to -1 . The ridge has a non-negligible curvature-driven terminal normal velocity.

This is the precise sense in which the relevant approximating manifold is a moving object. The curves $t \mapsto \Gamma_t(y)$ may be viewed as pursuit curves approaching the terminal manifold: they combine the large-scale dilation ty with the curvature correction of the Gaussian ridge.

Remark 4.3 (The ridge is not necessarily a material surface). The ridge $\mathcal{R}_t$ is an Eulerian feature of the density p_t . It need not be exactly transported by the generative drift a_t . In deterministic flow matching this distinction is already visible: the velocity at a ridge point and the time derivative of the ridge graph differ by terms coming from the changing Gaussian smoothing scale. The ridge is nevertheless the natural geometric object for the score because it is where the dominant normal score vanishes.

4.4 Eigenstructure of the score drift near the manifold

Return to the linear drift

$$a_t(x) = \frac{x}{t} + c_t s_t(x), \quad \nabla a_t(x) = \frac{1}{t} I_D + c_t \nabla s_t(x).$$

Combining this formula with Proposition 4.1 gives the block structure of the symmetric Jacobian.

Proposition 4.4 (Tangent–normal block estimates for the manifold score drift). *Fix $L > 0$. On the tube $\|n\| \leq L\sigma_t$, for t close enough to 1,*

$$Q_t \text{sym} \nabla a_t Q_t \preceq \left(\frac{1}{t} - \frac{c_t}{\sigma_t^2} + C_L c_t \right) Q_t = (\eta_t + C_L c_t) Q_t, \quad (78)$$

$$P_t \text{sym} \nabla a_t P_t \preceq \left[\frac{1}{t} + C_L c_t \left(1 + \frac{\|n\|}{\sigma_t^2} \right) \right] P_t, \quad (79)$$

$$\|P_t(\nabla a_t)Q_t\| + \|Q_t(\nabla a_t)P_t\| \leq C_L c_t \left(1 + \frac{1}{\sigma_t} \right). \quad (80)$$

On the thinner ridge tube $\|n\| \leq L\sigma_t^2$, the tangential block improves to

$$P_t \text{sym} \nabla a_t P_t \preceq \left(\frac{1}{t} + C_L c_t \right) P_t. \quad (81)$$

Proof. The normal estimate follows from (71):

$$Q_t \nabla a_t Q_t = \frac{1}{t} Q_t + c_t Q_t \nabla s_t Q_t \preceq \left(\frac{1}{t} - \frac{c_t}{\sigma_t^2} + C_L c_t \right) Q_t.$$

Since $c_t/\sigma_t^2 = 1/t - \eta_t$, this equals $(\eta_t + C_L c_t)Q_t$. The tangential estimate follows from (72), the bound $\|A_{t,n}\| \lesssim \|n\|/(t\tau_M)$, and the fact that $(I - A_{t,n})^{-1}$ is uniformly bounded in a fixed tube of radius smaller than the reach. The cross estimate follows similarly from (73). On the ridge tube $\|n\| = O(\sigma_t^2)$, the curvature term $\sigma_t^{-2}A_{t,n}$ is order one, which gives (81). $\square$

For the two schedules, (78) yields the same dominant normal contraction:

$$\eta_t \sim -\frac{1}{1-t} \quad (t \uparrow 1).$$

More explicitly,

$$\eta_t = -\frac{1}{1-t} \quad \text{for vanilla flow matching,}$$

and

$$\eta_t = -\frac{1+t^2}{t(1-t^2)} \sim -\frac{1}{1-t} \quad \text{for the rescaled diffusion schedule.}$$

Thus the $(D-d)$ normal eigenvalues of $\text{sym} \nabla a_t$ are negative and of order $-(1-t)^{-1}$. The d tangent eigenvalues may be positive, because they include tangential density curvature, geometric curvature, and the expansion x/t ; however their singularity is much weaker:

$$\begin{aligned} \text{typical tube } \|n\| = O(\sigma_t) : \quad & \lambda_{\max}(P_t \text{sym} \nabla a_t P_t) \lesssim 1 + c_t/\sigma_t, \\ \text{ridge tube } \|n\| = O(\sigma_t^2) : \quad & \lambda_{\max}(P_t \text{sym} \nabla a_t P_t) \lesssim 1 + c_t. \end{aligned}$$

For vanilla flow matching, $c_t \asymp \sigma_t$, so the typical tangential bound is $O(1)$. For the rescaled diffusion schedule, $c_t \asymp 1$, so the typical tangential bound is $O(\sigma_t^{-1}) = O((1-t)^{-1/2})$, still integrable.

4.5 Anisotropic manifold stability

We can now state the manifold version of the anisotropic estimate. Let $P_t(x)$ and $Q_t(x)$ denote tangent and normal projectors, either to M_t through the nearest-point projection, or more sharply to the ridge $\mathcal{R}_t$. Using the ridge projectors removes some order-one curvature bias but does not change the dominant normal contraction.

Theorem 4.5 (Manifold anisotropic stability, schematic sharp form). *Assume the manifold hypotheses above, and consider a learnt drift $\hat{a}_t$ whose symmetric Jacobian has, in the relevant high-probability tube around the moving ridge $\mathcal{R}_t$, the same block structure as the exact drift:*

$$\begin{aligned} P_t \text{sym} \nabla \hat{a}_t P_t &\preceq \ell_t^\parallel P_t, \\ Q_t \text{sym} \nabla \hat{a}_t Q_t &\preceq -\gamma_t^\perp Q_t, \end{aligned}$$

with moving-frame/off-diagonal coefficient κ_t , where near $t = 1$

$$\gamma_t^\perp \geq \frac{c_0}{1-t} - C, \quad \ell_t^\parallel \leq \begin{cases} C, & \text{vanilla flow matching in the typical tube,} \\ C(1-t)^{-1/2}, & \text{rescaled diffusion in the typical tube,} \\ C, & \text{either schedule in the ridge tube,} \end{cases} \quad (82)$$

and $\kappa_t \leq C(1 + c_t/\sigma_t)$. Then, for every terminal time $T < 1$,

$$W_1(p_T, \hat{p}_T) \leq \int_0^T \int (1, 1) R_{t,T} \begin{pmatrix} \|P_t(x)(a_t - \hat{a}_t)(x)\| \\ \|Q_t(x)(a_t - \hat{a}_t)(x)\| \end{pmatrix} p_t(dx) dt + \text{tube-tail errors}, \quad (83)$$

where $R_{t,T}$ is the 2×2 propagator in (51). In the ideal block-invariant case $\kappa = 0$, this becomes

$$\begin{aligned} W_1(p_T, \hat{p}_T) &\leq \int_0^T \exp\left(\int_t^T \ell_s^\parallel ds\right) \|P_t(a_t - \hat{a}_t)\|_{L^1(p_t)} dt \\ &\quad + \int_0^T \exp\left(-\int_t^T \gamma_s^\perp ds\right) \|Q_t(a_t - \hat{a}_t)\|_{L^1(p_t)} dt + \text{tube-tail errors}. \end{aligned} \quad (84)$$

In particular, since $\gamma_s^\perp \gtrsim (1-s)^{-1}$, the normal amplification satisfies

$$\exp\left(-\int_t^T \gamma_s^\perp ds\right) \lesssim \left(\frac{1-T}{1-t}\right)^{c_0} \quad (85)$$

up to harmless bounded factors. Normal errors are therefore damped as $T \uparrow 1$, whereas tangential errors are only multiplied by an integrable factor.

Proof. Apply Proposition 2.4 with the tangent and normal projectors associated with M_t or $\mathcal{R}_t$. The block estimates (82) are exactly those obtained in Proposition 4.4, with the additional assumption that the learnt drift obeys the same structural bounds. The tube-tail errors come from restricting the argument to the tube on which the nearest-point projection and the block estimates are valid; they are exponentially small in the chosen tube radius for the Gaussian interpolant. When $\kappa = 0$, the two-dimensional propagator is diagonal and gives (84). Finally, (85) follows by integrating $(1-s)^{-1}$:

$$\int_t^T \frac{ds}{1-s} = \log\left(\frac{1-t}{1-T}\right).$$

□

4.6 Consequences for score-aligned errors

The anisotropic estimate is useful only if a non-negligible part of the drift error lies in directions that are actually contracted by the terminal dynamics. Thus the slogan “score errors are normal errors” should not be used without qualification. The correct statement is local, terminal, and expressed in ridge coordinates. It is a statement about the structure of the score, not about changing the empirical loss used to learn it.

Let $q = D - d$ be the codimension. In the terminal tube we write

$$x = \Gamma_t(y) + \zeta, \quad \Gamma_t(y) = ty + \frac{\sigma_t^2}{2t} H_M(y) + O(\sigma_t^3), \quad \zeta \in N_y M, \quad (86)$$

so that $\Gamma_t(M) = \mathcal{R}_t$ is the moving ridge and ζ is the displacement from the ridge, not from the raw scaled manifold $M_t = tM$. In these coordinates, Proposition 4.1 becomes

$$Q_y s_t(\Gamma_t(y) + \zeta) = -\frac{\zeta}{\sigma_t^2} + R_t^\perp(y, \zeta), \quad P_y s_t(\Gamma_t(y) + \zeta) = \tau_t^\rho(y) + R_t^\parallel(y, \zeta), \quad (87)$$

where τ_t^ρ is the intrinsic tangential score generated by the density along M . In the smooth case,

$$\tau_t^\rho(y) = \frac{1}{t} \nabla_M \log \rho(y) + O(\sigma_t^\eta), \quad (88)$$

with $\eta > 0$ determined by the available regularity of M and ρ . The essential point is the separation of scales. On the typical shell $\|\zeta\| \asymp \sigma_t$,

$$\|Q_y s_t(x)\| \asymp \sigma_t^{-1}, \quad \|P_y s_t(x)\| \lesssim \|\tau_t^\rho\| + \|R_t^\parallel\|. \quad (89)$$

Thus the true score is mostly normal whenever

$$\sigma_t \left(\|\tau_t^\rho\| + \|R_t^\parallel\| \right) \ll 1. \quad (90)$$

For a fixed smooth positive density on a fixed smooth positive-reach manifold, this condition holds automatically as $\sigma_t \rightarrow 0$. It may fail, or at least become less informative, when the density oscillates at scale σ_t , when the tube radius is comparable to the reach, or when the score error is dominated by a tangential phase error.

A score-type drift error has the form

$$\delta a_t = c_t \delta s_t.$$

The normal damping in Theorem 4.5 is useful only for the normal part of δs_t . A convenient sufficient condition is a relative alignment model

$$\delta s_t(x) = \alpha_t(x) s_t(x) + r_t(x), \quad \|P_y r_t(x)\| \leq \varepsilon_t(x) \|Q_y \delta s_t(x)\|, \quad (91)$$

with no near-cancellation in the normal component. Combining (90) and (91) yields

$$\frac{\|P_y \delta s_t(x)\|}{\|Q_y \delta s_t(x)\|} \lesssim \sigma_t \left(\|\tau_t^\rho\| + \|R_t^\parallel\| \right) + \varepsilon_t(x). \quad (92)$$

In that regime the scalar one-sided Lipschitz estimate is pessimistic: it propagates the error as if it were tangent, while the error is mostly contained in the contracting normal block.

This alignment is plausible when the dominant error is a scalar error in the denoising strength, in the noise level, or in the magnitude of the normal restoring force. It is not the right model when the dominant error comes from estimating the density ρ along the manifold, from placing mass at the wrong tangential location on the ridge, or from an inaccurate parametrization of

the ridge. Those errors have a persistent tangential part, and the anisotropic stability bound correctly refuses to damp them away.

Curvature should also be read in this way. Its main role is not to create a leading tangential score. Its first-order effect is to move the ridge,

$$\mathcal{R}_t = \left\{ ty + \frac{\sigma_t^2}{2t} H_M(y) + O(\sigma_t^3) : y \in M \right\}, \quad (93)$$

and to rotate tangent and normal spaces. These effects remain lower order as long as

$$\sigma_t \|II_M\|_\infty \ll 1, \quad (94)$$

or, invariantly, as long as the Gaussian tube is thin compared with the reach. If σ_t is comparable to the reach, the tangent–normal splitting is itself unstable and the clean anisotropic picture should not be trusted.

5 Intrinsic anisotropic regularity, not a warped loss

The preceding discussion should not be interpreted as a proposal to modify the denoising score matching loss, or the empirical loss used in generator matching. The loss is fixed by the learning procedure. What the stability theorem provides is a *stability certificate*: a deterministic quantity which, if small, implies a small terminal sampling error. The question is then whether ordinary score estimation, together with the geometry of the target, naturally makes this certificate small.

The answer is that the manifold score has an intrinsic anisotropic regularity. In ambient coordinates the score is badly behaved because

$$Q_t \nabla s_t Q_t \simeq -\sigma_t^{-2} Q_t.$$

But this singularity is completely explicit: it is just the derivative of the normal affine restoring field. After subtracting this universal field and writing the score in ridge coordinates, the remaining object has the regularity of the density and of the manifold. The anisotropic quantity that appears in stability is therefore not a new loss. It is a proof quantity that becomes small because the target score has a structured tubular expansion.

5.1 Hölder geometry and density assumptions

We record the regularity scale explicitly. Let $0 < \alpha \leq 1$ and $\beta > 0$. We say that (M, ρ) belongs to the class

$$\mathfrak{M}_\tau^{k,\alpha}(L_M) \times \mathfrak{D}_{m,M}^\beta(L_\rho)$$

if the following conditions hold.

- (i) $M \subset \mathbb{R}^D$ is compact, embedded, without boundary, has reach at least $\tau > 0$, and admits $C^{k,\alpha}$ charts with norm at most L_M . For the ridge expansion one should take $k \geq 2$; for differentiated block estimates one should take $k \geq 3$.
- (ii) The density satisfies

$$0 < m \leq \rho \leq M_\rho < \infty, \quad \|\rho\|_{C^\beta(M)} \leq L_\rho.$$

If $\beta > 1$, then $\nabla_M \log \rho$ is a classical $C^{\beta-1}$ vector field. If $0 < \beta \leq 1$, the unsmoothed intrinsic score is not defined pointwise, but the Gaussian smoothing still produces a tangential score of size $\sigma_t^{\beta-1}$.

The exponent α controls the regularity of curvature. For instance, if $M \in C^{2,\alpha}$, then the mean-curvature vector is C^α in local coordinates, and the ridge graph

$$\Gamma_t(y) - ty = \frac{\sigma_t^2}{2t} H_M(y) + O(\sigma_t^{2+\alpha})$$

should be viewed as a C^α normal graph, up to higher-order terms. If $M \in C^{3,\alpha}$, then the projectors and the curvature corrections can be differentiated once more, which is the regularity level naturally used for the Jacobian estimates.

5.2 A regularized intrinsic tangential score

For low density regularity it is better not to force the expression $\nabla_M \log \rho$. Let $h_t = \sigma_t/t$, and denote by $H_{h_t^2}^M$ the heat semigroup on M at time comparable to h_t^2 . The tangential part of the Gaussian-convolved score is, up to curvature errors, the regularized intrinsic score

$$\tau_{t,\sigma}^\rho(y) := \frac{1}{t} \nabla_M \log(H_{h_t^2}^M \rho)(y). \quad (95)$$

This notation is useful because it covers both smooth and merely Hölder densities. Standard heat-kernel Schauder estimates give, schematically,

$$\|\tau_{t,\sigma}^\rho\|_{L^\infty(M)} \leq Ct^{-1} \begin{cases} \|\nabla_M \log \rho\|_{L^\infty(M)} + h_t^{\beta-1} [\nabla_M \log \rho]_{C^{\beta-1}}, & \beta > 1, \\ h_t^{\beta-1} \rho_{\min}^{-1}[\rho]_{C^\beta(M)}, & 0 < \beta \leq 1, \end{cases} \quad (96)$$

and, when $\beta > 1$,

$$\left\| \tau_{t,\sigma}^\rho - \frac{1}{t} \nabla_M \log \rho \right\|_{L^\infty(M)} \leq Ct^{-1} h_t^{\beta-1} [\nabla_M \log \rho]_{C^{\beta-1}}. \quad (97)$$

The case $0 < \beta \leq 1$ is not pathological: the tangential score may blow up like $\sigma_t^{\beta-1}$, but this is still less singular than the normal score σ_t^{-1} . In particular, on a typical normal shell, the ratio between tangential and normal sizes is at most of order σ_t^β , up to curvature constants.

5.3 Anisotropic Hölder expansion of the score

Use ridge coordinates

$$x = \Gamma_t(y) + \sigma_t \nu, \quad \nu \in N_y M, \quad \|\nu\| \leq L. \quad (98)$$

Define the residual

$$R_t(y, \nu) := s_t(\Gamma_t(y) + \sigma_t \nu) + \frac{\nu}{\sigma_t} - \tau_{t,\sigma}^\rho(y). \quad (99)$$

The point of this definition is that the singular part of the score has been removed before measuring regularity. The natural estimate is then anisotropic in the variables (y, ν) : derivatives in the ambient normal coordinate cost powers of σ_t^{-1} , but derivatives in the scaled normal variable ν do not.

A schematic but useful form is the following. If $M \in C^{2,\alpha}$, $\rho \in C^\beta$, $\rho \geq m > 0$, and $\sigma_t \ll t\tau$, then for every fixed L ,

$$\|R_t\|_{C_y^\theta C_\nu^0(\|\nu\| \leq L)} + \|R_t\|_{C_y^0 C_\nu^\theta(\|\nu\| \leq L)} \leq C_L \left(\sigma_t^\theta + \frac{\sigma_t}{t\tau} + \sigma_t \|II_M\|_\infty^2 \right), \quad (100)$$

where

$$\theta := \min\{\alpha, \beta\} \quad \text{if } 0 < \beta \leq 1, \quad \theta := \min\{\alpha, \beta - 1\} \quad \text{if } \beta > 1 \quad (101)$$

when the leading tangential term is written as (95). If instead $\beta > 1$ and one subtracts the unsmoothed score $t^{-1}\nabla_M \log \rho$, then the same estimate holds with an additional approximation error of order $h_t^{\beta-1}$. Under the stronger assumption $M \in C^{3,\alpha}$, the same statement can be differentiated once, with the expected loss of one Hölder order.

The precise constants in (100) are not important for the stability discussion. What matters is that the remainder is controlled by the intrinsic Hölder norms of M and ρ , and not by the ambient singular quantity σ_t^{-2} . This is the clean mathematical sense in which the score has anisotropic regularity.

Remark 5.1 (Why this is not an anisotropic loss). Equation (100) is a regularity estimate on the target score. It does not prescribe how the score should be trained. The empirical objective may remain the usual Euclidean denoising score matching objective. The role of the anisotropic expansion is only to prove that certain components of the learnt score error enter the final sampling error with small stability weights, while other components remain persistent.

5.4 The stability certificate induced by the expansion

Let $\delta s_t = \widehat{s}_t - s_t$ and $\delta a_t = c_t \delta s_t$. The matrix stability theorem gives the deterministic implication

$$W_1(p_T, \widehat{p}_T) \leq \int_0^T \int c_t \left\langle \delta s_t(x), G_{t,T}^{\widehat{a}}(x) \delta s_t(x) \right\rangle^{1/2} p_t(dx) dt. \quad (102)$$

This right-hand side is a stability certificate. In the manifold regime, and ignoring tangent–normal mixing for readability, it is bounded by

$$\mathfrak{S}_T(\delta s) := \int_0^T c_t A_{t,T}^{\parallel} \|P_t \delta s_t\|_{L^1(p_t)} dt + \int_0^T c_t A_{t,T}^{\perp} \|Q_t \delta s_t\|_{L^1(p_t)} dt, \quad (103)$$

where

$$A_{t,T}^{\parallel} = \exp \left(\int_t^T \ell_s^{\parallel} ds \right), \quad A_{t,T}^{\perp} = \exp \left(- \int_t^T \gamma_s^{\perp} ds \right).$$

The normal factor satisfies $A_{t,T}^{\perp} \lesssim ((1-T)/(1-t))^{c_0}$, whereas the tangent factor is governed only by an integrable expansion rate.

Suppose now that the learnt score is represented in the same intrinsic coordinates,

$$\widehat{s}_t(\widehat{\Gamma}_t(y) + \sigma_t \nu) = -\frac{\nu}{\sigma_t} + \widehat{\tau}_t(y) + \widehat{R}_t(y, \nu). \quad (104)$$

Then, to first order,

$$\|P_t \delta s_t\|_{L^1(p_t)} \lesssim \|\widehat{\tau}_t - \tau_{t,\sigma}^{\rho}\|_{L^1(M)} + \|P_t(\widehat{R}_t - R_t)\|_{L^1} + \varepsilon_{\text{mix},t}, \quad (105)$$

$$\|Q_t \delta s_t\|_{L^1(p_t)} \lesssim \frac{\|Q(\widehat{\Gamma}_t - \Gamma_t)\|_{L^1(M)}}{\sigma_t^2} + \|Q_t(\widehat{R}_t - R_t)\|_{L^1} + \varepsilon_{\text{mix},t}. \quad (106)$$

The term $\sigma_t^{-2}\|Q(\widehat{\Gamma}_t - \Gamma_t)\|$ is unavoidable: a normal displacement of the ridge by Δ changes the leading normal score by Δ/σ_t^2 . Thus normal directions are not intrinsically easy in a naive Euclidean sense. What is easy is the functional form of the normal component: once the ridge is known at sufficient precision, the leading normal restoring field is explicit.

Combining (103) with (105)–(106) gives

$$\begin{aligned} \mathfrak{S}_T(\delta s) &\lesssim \int_0^T c_t A_{t,T}^{\parallel} \left(\|\widehat{\tau}_t - \tau_{t,\sigma}^{\rho}\|_{L^1(M)} + \varepsilon_{R,t}^{\parallel} + \varepsilon_{\text{mix},t} \right) dt \\ &\quad + \int_0^T c_t A_{t,T}^{\perp} \left(\frac{\varepsilon_{\Gamma,t}^{\perp}}{\sigma_t^2} + \varepsilon_{R,t}^{\perp} + \varepsilon_{\text{mix},t} \right) dt. \end{aligned} \quad (107)$$

This is the promised synthesis. The empirical loss is not warped. Instead, the intrinsic regularity of the target score suggests a decomposition for the error, and the stability theorem tells us which pieces of that decomposition actually matter for the final generator. The tangential density score is persistent; the normal amplitude error is dynamically damped; the ridge-location error is normal but geometrically delicate because of the factor σ_t^{-2} .

5.5 Density regularity, curvature, and expected difficulty

The density controls the persistent tangential component. If ρ is constant, then $\tau_{t,\sigma}^\rho = 0$, up to curvature and heat-kernel corrections. If $\rho \in C^\beta$ with $\beta > 1$, then the difficult tangential object is essentially $\nabla_M \log \rho \in C^{\beta-1}$, a d -dimensional statistical object. If $0 < \beta \leq 1$, then the smoothed tangential score has size $\sigma_t^{\beta-1}$: it can grow as $t \uparrow 1$, but it is still less singular than the normal score, and its contribution is governed by the tangential stability factor.

Curvature controls a different object: the ridge and the rotation of tangent–normal spaces. If $M \in C^{2,\alpha}$, the mean curvature is C^α , so the ridge correction $\sigma_t^2 H_M / (2t)$ is smoother only to order α . If $M \in C^{3,\alpha}$, the Jacobian-level quantities entering the tangent–normal block estimates have the corresponding C^α control. Curvature remains harmless as long as $\sigma_t \|II_M\|_\infty \ll 1$. When this fails, the moving ridge may still exist locally, but the clean block decomposition loses uniform constants.

The final message is therefore not that one should change the loss. It is that the score admits an intrinsic anisotropic approximation theory. The normal component has a universal affine leading term around a moving ridge; the tangential component is the intrinsic score of a d -dimensional density; the curvature contributes lower-order residuals and frame rotation. The matrix $\widehat{G}_{t,T}^a$ then converts this intrinsic approximation structure into a stability bound for the terminal generator.